

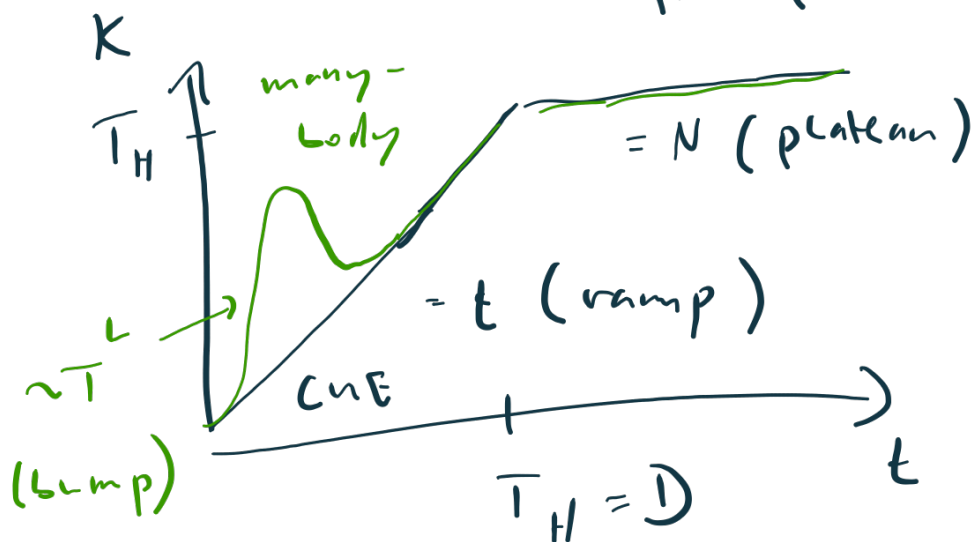
SFF from Semiclassics [arXiv: 2605.27052]

- From single particle to structured many-body (Floquet) systems

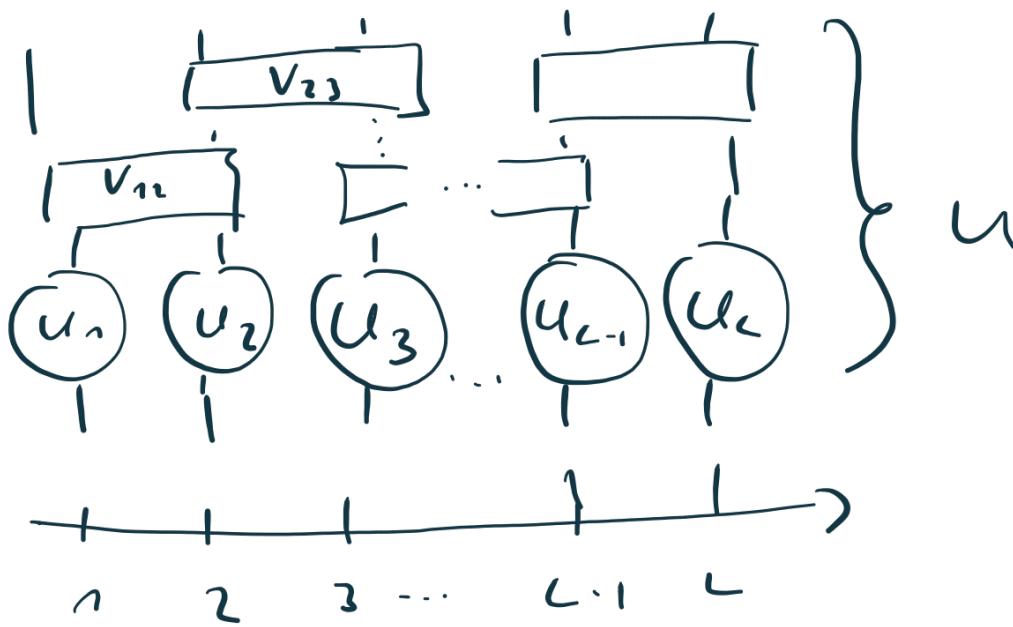
▮ SFF: Floquet operator U

$$K(t) = |\text{tr } U^t|^2 \quad (t > 0)$$

= FT (probability of finding two levels at distance ω)



Setting: (morally Random Phase Model)



quantum:

$$H_x = \text{span} \{ |q_x\rangle \}, \quad \dim H_x = N \rightarrow t = \frac{1}{2\pi N}$$

$$\langle q' | U_\varepsilon | q \rangle \propto \left| \det \frac{\partial^2 W_\varepsilon(q, q')}{\partial q \partial q'} \right|^{\frac{1}{2}} e^{\frac{i}{\varepsilon} W_\varepsilon(q, q')}$$

$W_\varepsilon(q, q')$... generating function

classical: ($t \rightarrow 0$)

compact phase space $M^L = M \times M \times \dots \times M$

with coordinates (q, p)

Symplectic Map: $(q', p') = \bar{\Phi}_\varepsilon(q, p)$

$$\text{s.t. } p_x = \frac{\partial W_\varepsilon}{\partial q_x}$$

$$p'_x = \frac{\partial W}{\partial q'_x}$$

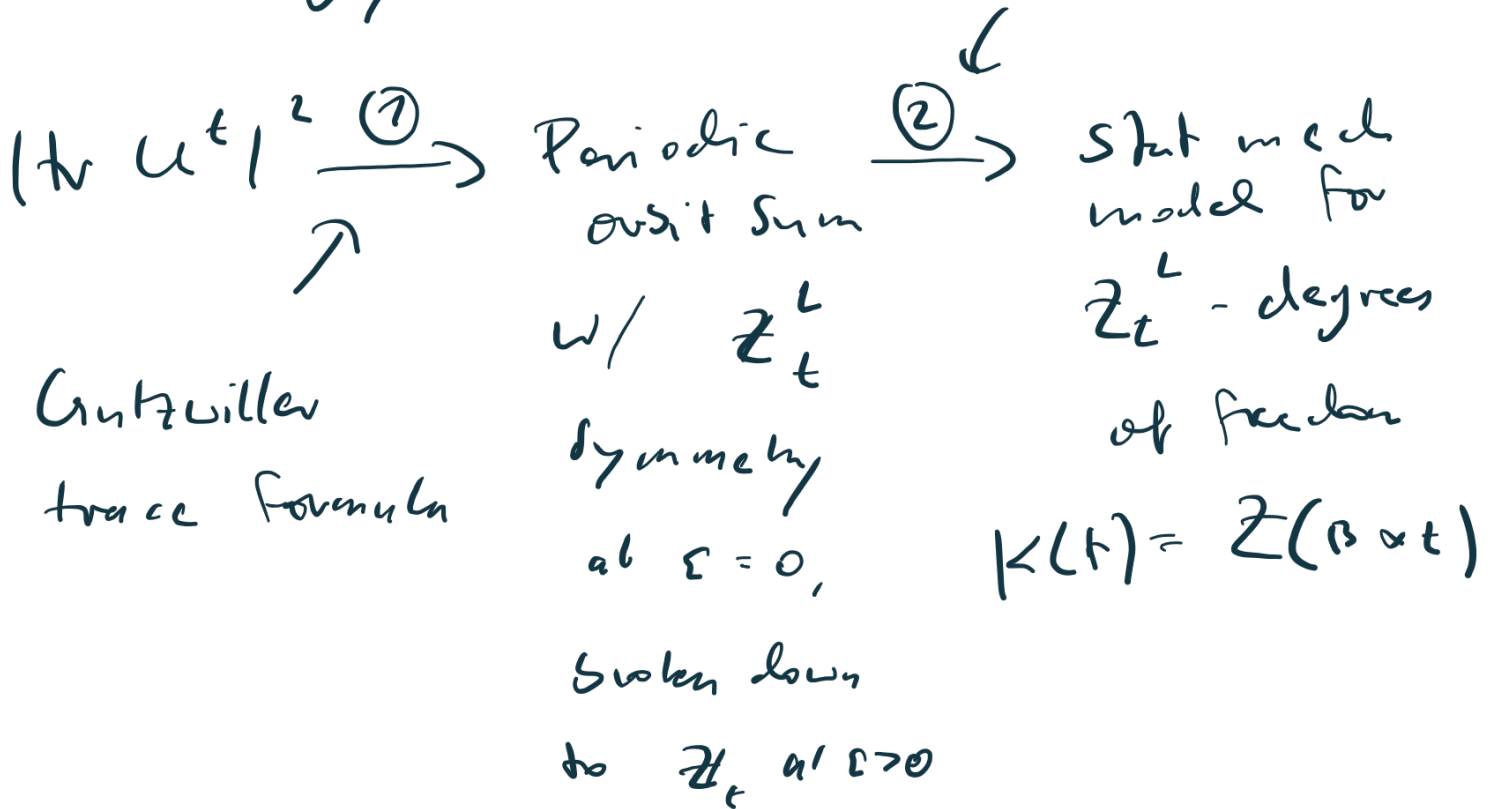
$$W_\varepsilon(q, q') = \sum_x W^{(x)}(q_x, q'_x) + \varepsilon V(q, q') + o(\varepsilon^2)$$

$$\Phi_{\varepsilon=0} = \mathcal{Q}_1 \times \mathcal{Q}_2 \times \dots \times \mathcal{Q}_L \rightarrow \text{uniformly hyperbolic}$$

$$U_{\varepsilon=0} = U_1 \otimes U_2 \otimes \dots \otimes U_L \rightarrow \text{quantum chaotic}$$

Strategy:

Sum rules + CLT



① Gutzwiller trace formula:

$$\text{tr}(U^t) = \int_{M^t} dq^{(0)} \dots dq^{(t-1)} K(\vec{q}) e^{\frac{i}{\hbar} S(\vec{q})}$$

$$S(\vec{q}) = \sum_{i=0}^{t-1} W(q^{(i)}, q^{(i+1)}) \quad q^{(t)} = q^{(0)}$$

Stationary phase approximation:

$\vec{q}$ = position along a t -periodic orbit

$$\{ \Phi^s(q^{(0)}, p^{(0)}) \}_{s=0, \dots, t}$$

p.o. parametrized by initial condition y

$$\Rightarrow \text{tr}(U^t) = \sum_{y: \Phi^t(y) = y} A_y e^{\frac{i}{\hbar} S_y}$$

↓
stability of orbit

↓
classical action of the orbit

(the less unstable the larger the contribution)

$\Rightarrow$ SFT

$$K(t) = \sum_{\gamma, \gamma'} A_{\gamma} A_{\gamma'}^* e^{\frac{i}{t}(S_{\gamma} - S_{\gamma'})}$$

as $t \rightarrow 0$ only $\gamma, \gamma' \wedge S_{\gamma} = S_{\gamma'}$

$$\Leftrightarrow \gamma' = \Phi^s \gamma, s \in \mathbb{Z}_t \quad (\Rightarrow A_{\gamma} = A_{\gamma'})$$

("Diagonal approximation")

$$\Rightarrow \underline{K(t)} = \sum_{s \in \mathbb{Z}_t} \sum_{\gamma} |A_{\gamma}|^2 \stackrel{(*)}{=} \sum_{s \in \mathbb{Z}_t} 1 = \underline{t}$$

$= RMT$

τ (2.A)

Birkhoff's ergodic theorem

$$\frac{1}{T} \sum_{s=0}^{T-1} \delta_{\Phi^s(x)} \longrightarrow \mu \text{ (Liouville (SRB) measure)}$$

for almost all x

• if γ t -periodic: $\frac{1}{t} \sum_{s=0}^{t-1} \delta_{\Phi^s(\gamma)} = \delta_{\gamma}^{(t)} \neq \mu$

But $\sum_{\gamma} |A_{\gamma}|^2 \delta_{\gamma}^{(t)} \xrightarrow{t \rightarrow \infty} \mu$

Integrating const. Fct. 1 over M^L
w.r.t. both measures yields (v)

L = Hannay - Ozorio de Almeida sym rule
 RMT - SFF for single particle systems
 and many-body systems at fixed ε .

⇒ Scaling Limit: $\varepsilon \rightarrow 0, \hbar \rightarrow 0$

$$\text{but } \Lambda = \varepsilon^2 T_H / \tau^2 \propto \varepsilon^2 / \hbar^{2+2} = \text{const.}$$

→ non-trivial deviations from RMT SFF

• Linearize $S_y - S_y'$ around $\varepsilon = 0$

and use periodic orbits of $\bar{\Phi}_{\varepsilon=0}$

• systematic contributions only if

$$x' = \bar{\Phi}_0^{\vec{s}} x = (\varphi_1^{s_1} x_1, \dots, \varphi_L^{s_L} x_L)$$

$$\vec{s} \in \mathbb{Z}_t^L$$

("Diagonal approximation")

→ Families of periodic orbits

parametrized by $\mathbb{Z}_t^L$ (discrete torus)

$$\frac{1}{\epsilon} (S_T - S_{T'}) = \sqrt{\Delta\tau} \Delta_{\vec{s}}(y) + \text{higher order terms which vanish as } \epsilon, \tau \rightarrow 0$$

$$t + \vec{s} = t \vec{1} + \vec{s} = t(1, 1, \dots, 1) + \vec{s}$$

$$\Delta_{\vec{s}}(y) = \frac{1}{\sqrt{\epsilon}} \sum_{r=0}^{t-1} \underbrace{V(\Phi_0^t y, \Phi_0^{t+1} y) - V(\Phi_0^{t+\vec{s}} y, \Phi_0^{t+\vec{s}+1} y)}_{\downarrow}$$

$$= \frac{1}{\sqrt{\epsilon}} \sum_{r=0}^{t-1} V_{\vec{s}}(y)$$

$$\Rightarrow K(t) = \sum_{\vec{s} \in \mathcal{H}_t^L} \sum_y |A_y|^2 e^{i\sqrt{\Delta\tau} \Delta_{\vec{s}}(y)}$$

• sanity check: $V=0 \Rightarrow \Delta_{\vec{s}}(y)=0$

$$\Rightarrow K(t) = \sum_{\vec{s} \in \mathcal{H}_t^L} 1 = t^L$$

... RPT SFF for L independent

ChC / non-interacting chaotic systems

⌈ (2B) fluctuations of

$$F(t) = \frac{1}{\epsilon} \sum_{j=0}^{t-1} f(\bar{\Phi}^j(x)) \text{ around } \int f d\mu (=0)$$

(amplified alk rescaling by $\sqrt{t}$)

are Gaussian and obey CLT

$$e^{i\sqrt{\Lambda t} F(t)} \sim e^{-\Lambda t \frac{\sigma_t^2}{2t}}$$

(Fourier-version of CLT)

CLT
 $\Rightarrow$
 H: O.d.I sum rule

$$K(t) = \sum_{\vec{j} \in \mathcal{H}_t^L} e^{-\Lambda t \frac{\sigma_{\vec{j}}^2}{2}} \quad \begin{matrix} \downarrow & \downarrow \\ \beta & H(\vec{j}) \end{matrix}$$

partition function $Z(\beta)$

$$\frac{\sigma_{\vec{j}}^2}{2} = H(\vec{j}) = \sum_{\epsilon \in \mathcal{H}} \int d\mu \underbrace{v_{\vec{j}}(\bar{\Phi}_0^\epsilon(x)) v_{\vec{j}}(x) d\mu(x)}_{\text{2-point correlation function of } v_{\vec{j}}}$$

2-point correlation function of $v_{\vec{j}}$

$$\Rightarrow H(\vec{s}) \geq 0$$

$$H(\vec{s}) = 0 \quad \text{i.t.f.} \quad \vec{s} = s' \vec{1}$$

(under mild assumptions)

$\rightarrow$ ϵ - Fold degenerate ground state $\rightarrow$ RMT SFF

$$H(\vec{s} + s' \vec{1}) = H(\vec{s})$$

V local (n.n. coupling)

$\Rightarrow H$ local (n.n. interaction)

instant decay of correlations for V

$$H(\vec{s}) = \begin{cases} 0 & s \propto \vec{1} \\ h_0 = \text{const.} & \text{otherwise} \end{cases} \quad (\text{I})$$

instant decay of correlations + local (II)

$H(\vec{s}) \dots$ t-states Potb model

$$(\text{I}) \quad K(t) = t + (t^L - t) e^{-\Lambda \tau h_0}$$

$\uparrow$
RMT SFF

(from diagonal shifts)

$\rightarrow 0$ as $t \rightarrow \infty$

(from off-diagonal shifts $\rightarrow$ dephasing)

$$(II) \quad K(t) = (1 - e^{-\Lambda \tau \hat{h}_0} + t e^{-\Lambda \tau \hat{h}_0})^L$$

$$= \text{RPM} \quad \checkmark \quad + (t-1) (1 - e^{-\Lambda \tau \hat{h}_0})^L$$

$\searrow \quad \checkmark$
 RMT SFR

(III) subexponential decay of correlations (mixing)

$$|K_{III}(t) - K_I(t)| \leq C t^{L+1} e^{-ct} \quad (C, c > 0)$$

(I, II, III): Bump - Ramp structure,

no plateau:

Semiclassics valid For

$$1 \ll t \ll N = T_H^{1/2}$$